QUESTION 1

Which of the following best describes an analog signal?

Answer is Wrong

Score: 0 / 1

Options

A. A signal that can take only two values

B. A signal that switches between binary states

Expected

C. A continuously variable signal

Wrong

D. A signal always in digital format

FEEDBACK

Solution: Analog signals can take on a continuous range of values, unlike digital signals which are binary.

Link: https://youtu.be/76fR60p294g?feature=shared&t=26

QUESTION 2

Why must analog signals be converted before a microcontroller can process them?

Not Attempted **Score: 0 / 1**

Options

Expected

A. Microcontrollers cannot store real-valued numbers directly

B. Analog signals are too fast for microcontrollers

C. Binary numbers consume less memory

D. To reduce electromagnetic interference

FEEDBACK

Solution: The CPU in a microcontroller processes only binary numbers, so analog signals must be converted into digital form first.

Link: https://youtu.be/76fR60p294g?feature=shared&t=158

QUESTION 3

What is the process of recording an analog signal's value at specific time intervals called?

Not Attempted **Score: 0 / 1**

Options

A. Quantization

Expected

B. Sampling

C. Filtering

D. Resolution

FEEDBACK

Solution: Sampling means taking an analog signal and storing its value at discrete time intervals.

Link: https://youtu.be/76fR60p294g?feature=shared&t=205

QUESTION 4

In ADCs, discretizing both time and amplitude produces:

Not Attempted **Score: 0 / 1**

Options

A. A quantized analog waveform

B. A binary logic signal

Expected

C. A digital signal

D. A continuous-time waveform

FEEDBACK

Solution: Discretizing in both time and amplitude transforms the analog waveform into a digital signal.

Link: https://youtu.be/76fR60p294g?feature=shared&t=453

QUESTION 5

What does the “full-scale range” of an ADC refer to?

Not Attempted **Score: 0 / 1**

Options

A. The maximum current the ADC can handle

B. The maximum binary number it can output

Expected

C. The range of input voltages it can convert

D. The highest frequency it can sample

FEEDBACK

Solution: The full-scale range is the span of input voltages over which the ADC can operate.

Link: https://youtu.be/76fR60p294g?feature=shared&t=496

QUESTION 6

If an ADC has 10-bit resolution and a 0–3.3V range, what is the step size?

Not Attempted **Score: 0 / 1**

Options

A. 3.3 V

Expected

B. 3 mV

C. 0.33 V

D. 1 mV

FEEDBACK

Solution: Step size = (Vmax – Vmin) / (2^n) = 3.3 / 1024 ≈ 3 mV.

Link: https://youtu.be/76fR60p294g?feature=shared&t=738

QUESTION 7

Which factors can limit the effective resolution of an ADC despite a high bit count?

Not Attempted **Score: 0 / 1**

Options

A. Physical noise in the system

B. Nonlinearities in the ADC

C. Noise in amplifiers

Expected

D. All of the above

FEEDBACK

Solution: Noise from the system, amplifiers, and ADC imperfections can reduce effective resolution.

Link: https://youtu.be/76fR60p294g?feature=shared&t=792

QUESTION 8

According to the Nyquist Criterion, what is the minimum sampling rate needed for accurate reconstruction?

Not Attempted **Score: 0 / 1**

Options

A. Equal to the signal's highest frequency

Expected

B. Twice the signal's highest frequency

C. Half the signal's highest frequency

D. Dependent only on ADC resolution

FEEDBACK

Solution: The Nyquist Criterion states the sampling frequency must be at least twice the highest frequency component of the signal.

Link: https://youtu.be/76fR60p294g?feature=shared&t=919

QUESTION 9

Why might you choose an 8-bit ADC setting instead of 12-bit?

Not Attempted **Score: 0 / 1**

Options

A. 8-bit ADCs are always more accurate

Expected

B. Faster conversion and less power usage

C. 8-bit ADCs support more channels

D. 12-bit ADCs cannot sample continuously

FEEDBACK

Solution: Lower resolution allows faster conversion, lower power consumption, and less memory usage.

Link: https://youtu.be/76fR60p294g?feature=shared&t=1123

QUESTION 10

What is a limitation of using a single ADC for multiple channels?

Not Attempted **Score: 0 / 1**

Options

Expected

A. Only one channel can be sampled at a time

B. Lower resolution per channel

C. Increased input voltage range

D. No support for continuous sampling

FEEDBACK

Solution: Multiplexing means channels are sampled sequentially, not simultaneously.

Link: https://youtu.be/76fR60p294g?feature=shared&t=1296

QUESTION 11

What is the function of a sample-and-hold circuit in an ADC?

Not Attempted **Score: 0 / 1**

Options

A. To store digital output temporarily

Expected

B. To hold the input voltage constant during conversion

C. To reduce power consumption

D. To synchronize multiple ADCs

FEEDBACK

Solution: The sample-and-hold circuit captures and maintains the input voltage so the ADC can convert it accurately.

Link: https://youtu.be/76fR60p294g?feature=shared&t=1473

QUESTION 12

In the Arduino example, what type of ADC triggering is used with analogRead()?

Not Attempted **Score: 0 / 1**

Options

A. Continuous hardware triggering

Expected

B. Software-triggered conversion

C. Timer-based automatic triggering

D. Event-based sampling only

FEEDBACK

Solution: analogRead() starts the ADC conversion in software only when the function is called.

Link: https://youtu.be/76fR60p294g?feature=shared&t=1933